Excercises

Problem 1

Suppose that the data for analysis includes the attribute age. The age values for the data tuples are (in increasing order) {13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70}.

a. What is the *mean* of the data? What is the *median*?
b. What is the *mode* of the data? Comment on the data's modality (i.e., bimodal, trimodal, etc.).
c. What is the *midrange* of the data?
d. Can you find (roughly) the first quartile (Q_1) and the third quartile (Q_3) of the data?
e. Give the *five–number summary* of the data.
f. Show a *boxplot* of the data.
g. What is the *variance* and *standard deviation* of the data?

Problem 2

Suppose that a hospital tested the age and body fat data for 18 randomly selected adults with the following results:

age	23	23	27	27	39	41	47	49	50
%fat	9.5	26.5	7.8	17.8	31.4	25.9	27.4	27.2	31.2
age	52	54	54	56	57	58	58	60	61
%fat	34.6	42.5	28.8	33.4	30.2	34.1	32.9	41.2	35.7

a. Calculate the mean, median, variance, and standard deviation of *age* and *%fat*.
b. Draw the boxplots for *age* and *%fat*.
c. Draw a *scatter plot* for these two variables.
d. Calculate the Pearson and Spearman correlation coefficients between the two attributes.

Problem 3

Let D be the number of appearances, goals, and assists of Cristiano Ronaldo in Champions League from season 03/04 to 21/22.

a. Calculate the *mean*, *median*, *mode*, *variance*, *standard deviation* of three attributes appearances, goals, and assists.
b. Find the *first quartile* (Q_1) and the *third quartile* (Q_3) of three three attributes appearances, goals, and assists.
c. Show the *boxplots* of the three attributes appearances, goals, and assists.
d. Show the *scatter plots* of appearances and goals; of appearances and assists; of goals and assists.
e. Compute the *covariance matrix* of the three attributes appearances, goals, and assists.

Excercises (cont'd)

$\mathcal{D} =$

Club	*Season*	*Appearances*	*Goals*	*Assists*
Man United	03/04	5	0	1
Man United	04/05	7	0	2
Man United	05/06	6	0	0
Man United	06/07	11	3	5
Man United	07/08	11	8	1
Man United	08/09	12	4	3
Real Madrid	09/10	6	7	1
Real Madrid	10/11	12	6	4
Real Madrid	11/12	10	10	3
Real Madrid	12/13	12	12	1
Real Madrid	13/14	11	17	5
Real Madrid	14/15	12	10	4
Real Madrid	15/16	12	16	4
Real Madrid	16/17	13	12	6
Real Madrid	17/18	13	15	3
Juventus	18/19	9	6	2
Juventus	19/20	8	4	1
Juventus	20/21	6	4	2
Man United	21/22	7	6	0

Excercises (cont'd)

Problem 4

800 students of three majors (math, history, and computer science) were asked whether they like playing computer games or not. The survey data are summarized in the following contingency table:

	Like games	Do not like games	Total
Math	130	100	230
History	35	165	200
Computer science	280	90	370
Total	445	355	800

a. Calculate the expected frequency of each cell in the table above.

b. Using Pearson's chi–square test to confirm that there is a correlation between study major and playing computer games or not?